

Mohit Kumar

Senior Product Design Engineer | New Product Development (NPD) |
Sheet Metal & Furniture Design
Narela, Delhi – 110040
9311301814 | mohitgaigan@gmail.com
 <https://mohitgaigan.github.io/cad-design-lab/>

About Me

Product Design Engineer with nearly 5 years of experience in New Product Development (NPD) specialising in sheet metal, modular furniture, laboratory furniture, and office furniture. Designed 300+ products and contributed to 50+ commercial projects valued up to ₹50 lakh. Experienced in DFM, engineering drawings, BOM preparation, product costing, material selection, and manufacturing support. Reduced manufacturing costs by approximately 20% through design optimisation while improving product functionality.

Work Experience

- DELITE KOM LIMITED** Sep 2023 - Present
Product Design Engineer
 - Led end-to-end New Product Development (NPD) for modular, laboratory, and office furniture.
 - Designed 300+ production-ready products using AutoCAD, SolidWorks, and 3ds Max.
 - Delivered 50+ commercial projects valued up to ₹50 lakh from concept to installation.
 - Prepared engineering drawings, fabrication drawings, assembly drawings, BOMs, quotations, and product costing.
 - Reduced manufacturing costs by approximately 20% through design optimisation, value engineering, and material selection.
 - Developed photorealistic 3D models, rendered layouts, and space-planning visualisations to support client presentations and accelerate design approvals.
 - Prepared product cost estimates and quotations while coordinating with procurement and production teams.
 - Collaborated with production, procurement, fabrication, and installation teams to ensure manufacturability.
 - Trained Junior Designers and Quality Engineers while supporting client meetings and technical reviews.
 - Served as ISO Coordinator for ISO 9001, ISO 14001, and ISO 45001 documentation and internal audits.
- TECH MAHINDRA** JAN 2021 - 2022
Design Engineer
 - Designed mechanical and electrical components using AutoCAD in accordance with engineering and manufacturing standards.
 - Supported New Product Introduction (NPI) activities by preparing engineering drawings and technical documentation.
 - Identified equipment specifications for new products to meet quality and cost requirements.
 - Worked with design and procurement teams for equipment selection.
 - Coordinated the introduction of new products onto the manufacturing floor.
 - Provided technical support to manufacturing during new product introduction.
 - Responsible for resolving problems; identifying and analysing changes in dimensions.
 - Trained two new employees in industry knowledge and the AutoCAD tool; new employee performance was notably higher than average.

Education

- Chandigarh University (Pursuing)
Master of Business Administration (MBA) Dual specialization: Operations Management & Digital Marketing
- Maharshi Dayanand University (MDU) 2017
Bachelor of Technology (B.Tech.) – Mechanical Engineering
- LOVELY PUBLIC SCHOOL 2012
Higher Secondary (CBSE)
- ST. LAWRENCE PUBLIC SCHOOL 2010
SECONDARY EDUCATION (CBSE)

Certifications

- SOLIDWORKS CERTIFIED PROFESSIONAL (CSWP) - RVM CAD
- AUTOCAD CERTIFIED PROFESSIONAL - RVM CAD
- POWERMILL CAM PROGRAMMER - RVM CAD
- 3ds Max, V-Ray & Lumion for Architectural Visualizatio - PTS CAD Expert

Technical Skills

- CAD & Engineering: AutoCAD, SolidWorks, Sheet Metal Design, Weldments, Assembly Modelling, Surface Modelling, GD&T, Engineering Drawings.
- Rendering & Visualisation: 3ds Max, V-Ray, Corona Renderer, Lumion, PhotoView 360.
- Manufacturing & CAM: PowerMill, Delcam, Product Costing, BOM Preparation, CNC Manufacturing, VMC Programming.
- Manufacturing Processes: Press Brake, MIG Welding, Spot Welding, Shearing, Power Press, Powder Coating, Pipe Cutting & Bending.
- Design & Productivity Tools: Adobe Photoshop, Adobe Illustrator, Canva, Microsoft Excel, Microsoft PowerPoint, Google Sheets.

Professional Competencies

- Product Development: NPD, DFM, Product Cost Optimisation, Material Selection
- Design Engineering: Hand Sketch Interpretation, Concept Development, AutoCAD 2D Drafting, 3D Product Modelling (SolidWorks), Assembly Modelling, Engineering Drawings, Fabrication Drawings, BOM Preparation, GD&T.
- Project Management: Client Coordination, Cross-functional Collaboration, Production Support, Installation Support.
- Experience converting hand sketches and AutoCAD drawings into accurate SOLIDWORKS 3D models.
- CAM & CNC Programming: PowerMill CAM Programming, NC Code Generation, G & M Code Programming, VMC Machining Support.
- Quality & Compliance: ISO 9001, ISO 14001, ISO 45001, Internal Audits.
- Effective teamwork, communication, and project coordination abilities.

Key Projects

- **PROJECT: ULTRATECH – OFFICE FURNITURE DESIGN AND LAYOUT**
 - Led the design and development of conference tables and modular workstations for the UltraTech project, from concept stage to final execution.
 - Managed the complete project design lifecycle, including space planning, product design, and client-specific customisation.
 - Designed sheet metal and wood-based conference tables and workstation systems, ensuring durability, ergonomics, and aesthetic alignment with corporate standards.
 - Prepared detailed technical drawings, fabrication drawings, and BOMs for smooth manufacturing and on-site execution.
 - Created 3D rendered floor plan layouts and photorealistic visuals to help stakeholders visualise the final office setup and approve designs efficiently.
 - Coordinated with production, fabrication, and installation teams to ensure design accuracy, manufacturability, and timely project delivery.
 - Ensured compliance with ergonomic norms, space utilisation requirements, and project-specific quality standards.
- **PROJECT: CFEES (CENTRE FOR FIRE, EXPLOSIVE AND FIRE SAFETY) CHEMISTRY LAB**
 - Led the design and development of chemistry laboratory furniture with a focus on functionality, safety, and efficient space utilisation.
 - Developed detailed 3D models and realistic rendered floor plan layouts to help clients clearly visualise laboratory setup, workflows, and final installations.
 - Designed sheet metal laboratory tables and storage units suitable for chemical lab environments, considering durability and maintenance requirements.
 - Coordinated closely with production, fabrication, and sourcing teams to ensure designs were feasible for sheet metal fabrication processes including bending, punching, welding, and powder coating.
 - Prepared technical drawings and fabrication-ready documentation to support accurate manufacturing and smooth on-site execution.
 - Ensured designs met project-specific standards, safety guidelines, and client requirements for laboratory environments.